SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

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1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

Given:

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
A – Viola-Jones	80	20	30
B – YOLO	90	15	20

(a) Compute Precision and Recall

Formulas

Precision measures how many of the detected objects are actually correct.

Precision=TP+FPTP

Recall measures how many of the actual objects were successfully detected.

Recall=TP+FNTP

Model A – Viola-Jones

Given:

- TP = 80
- FP = 20
- FN = 30

Precision

Precision=80+2080 =10080 =0.80=80%

This means **80% of the objects detected by Viola-Jones are actually correct**.

Recall

Recall=80+3080 =11080 =0.7273=72.73%

This means Viola-Jones detects about **72.73% of all actual objects**.

Model B – YOLO

Given:

- TP = 90
- FP = 15
- FN = 20

Precision

Precision=90+1590 =10590 =0.8571=85.71%

This means **85.71% of the objects detected by YOLO are actually correct**.

Recall

Recall=90+2090 =11090 =0.8182=81.82%
This means YOLO detects about **81.82% of all actual objects**.

Comparison

Model	Precision	Recall
Viola-Jones	80%	72.73%
YOLO	85.71%	81.82%

(b) Performance Evaluation
Precision:
YOLO has **85.71%**, while Viola-Jones has **80%**. Therefore, YOLO produces fewer false detections and gives more accurate object detections.
Recall:
YOLO has **81.82%**, while Viola-Jones has **72.73%**. Therefore, YOLO detects more of the actual objects and misses fewer objects.
True Positives:
YOLO detects **90 objects correctly**, compared with **80** for Viola-Jones.
False Positives:
YOLO has only **15 false positives**, compared with **20** for Viola-Jones.
False Negatives:
YOLO has **20 missed objects**, compared with **30** for Viola-Jones.
Conclusion
YOLO (Model B) is the better object detection model. ✅
It has **higher precision (85.71%) and higher recall (81.82%)** than Viola-Jones. Hence, YOLO provides **more accurate detections, fewer false positives, and fewer missed objects**, making it more effective for object detection.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object	Stereo Vision	Structure from Motion
A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4
(a) Compute absolute error for both methods.
(b) Compare and evaluate which method is more accurate.

Absolute Error Formula:
Absolute Error=∣Estimated Depth−Actual Depth∣
(a) Absolute Error

Object	Actual Depth	Stereo Vision	Stereo Error	SfM	SfM Error
A	2.0 m	2.0 m	**0 m**	2.5 m	**0.5 m**
B	3.0 m	3.0 m	**0 m**	3.8 m	**0.8 m**
C	4.0 m	4.0 m	**0 m**	5.0 m	**1.0 m**

(b) Comparison

- **Stereo Vision total error:** 0 + 0 + 0 = **0 m**
- **SfM total error:** 0.5 + 0.8 + 1.0 = **2.3 m**
- Average error:
  - Stereo = **0 m**
  - SfM = **0.77 m**

Conclusion

Stereo Vision is more accurate because its estimated depths exactly match the actual depths, giving **zero error**. SfM consistently overestimates the depth. ✅

7. **F1-Score Comparison**

Two models:

Model	**Precision**	**Recall**
A	0.8	0.6
B	0.75	0.75

(a) Compute F1-score for both models.
(b) Compare and evaluate which model is better.

(a) Compute F1-Score

Formula:

F1=Precision+Recall2×Precision×Recall

Model A

Precision = 0.8, Recall = 0.6

F1=0.8+0.62(0.8)(0.6) =1.40.96=0.686

F1-score = 68.6%

Model B

Precision = 0.75, Recall = 0.75

F1=0.75+0.752(0.75)(0.75) =1.51.125=0.75

F1-score = 75%

(b) Comparison

Model	Precision	Recall	F1-Score
A	0.80	0.60	**0.686**
B	0.75	0.75	**0.75**

Conclusion: Model **B is better** because it has a higher F1-score (**75%**) and provides a better balance between precision and recall. ✅

8. **Optical Flow Consistency**

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.
(b) Compare consistency and evaluate which is more stable.

(a) Average Motion

Method A:

Average=34+5+6=315=5

Method B:

Average=36+8+9=323=7.67

(b) Consistency Comparison

Method	Motion Values	Average
A	4, 5, 6	**5**
B	6, 8, 9	**7.67**

- Method A changes gradually: **4** → **5** → **6**
- Method B changes more irregularly: **6** → **8** → **9**
- Method A has lower variation, so it is **more consistent and stable**.

Conclusion

Method A is more stable because its motion values vary less between frames. ✅

9. **Detection Accuracy Comparison**

Results:

Model	Correct	Total
A	180	200
B	170	200

(a) Compute accuracy.
(b) Compare and evaluate.

(a) Compute Accuracy

Formula:

Accuracy=Total PredictionsCorrect Predictions×100

Model A:

Accuracy=200180×100=90%

Model B:

Accuracy=200170×100=85%

(b) Comparison

Model	Correct	Total	Accuracy
A	180	200	**90%**

Model	Correct	Total	Accuracy
B	170	200	**85%**

Conclusion
Model A is better because it has a higher accuracy of **90%**, compared with **85%** for Model B. Therefore, Model A gives **5% better accuracy**. ✅

10. **Error Rate Comparison**

Results:

Method	Errors	Total
A	20	100
B	10	100

(a) Compute error rate.
(b) Compare and evaluate.

(a) Compute Error Rate
Formula:
Error Rate=TotalErrors×100
Method A:
Error Rate=10020×100=20%
Method B:
Error Rate=10010×100=10%
(b) Comparison

Method	Errors	Total	Error Rate
A	20	100	**20%**
B	10	100	**10%**

Conclusion
Method B is better because it has a lower error rate of **10%**, compared with **20%** for Method A. Therefore, Method B is **more accurate and reliable**. ✅

11. **Motion Estimation Variance**

Errors:

Frame	A	B
1	2	6
2	3	7
3	2	5

(a) Compute mean error.
(b) Compare variability and evaluate stability.

(a) Compute Mean Error
Method A:
Mean=32+3+2=37=2.33

Method B:
Mean=36+7+5=318=6
(b) Compare Variability and Stability

Method	Errors	Mean Error
A	2, 3, 2	**2.33**
B	6, 7, 5	**6.00**

For stability, lower variation in errors is better. Method A's errors are closely grouped (**2, 3, 2**), while Method B has larger errors (**6, 7, 5**).
Var(X)=E(X2)−[E(X)]2
Var(X)=(1.4)2=1.96Var(X)=(\text{1.4})^2=\text{1.96}Var(X)=(1.4)2=1.96
σ\sigmaσ
σ\sigmaσ
μ-σ+σVar(X) ≈ 1.96
Give feedback
Conclusion
Method A is more stable and accurate because it has a much lower mean error and smaller variation in error values. ✅

12. **IOU Comparison**
IOU values:

Object		**A**	**B**
1	0.6	0.8	
2	0.7	0.85	
3	0.65	0.9	

(a) Compute average IOU.
(b) Compare and evaluate detection quality.

(a) Compute Average IoU
Formula:
Average IoU=Number of objectsSum of IoU values
Method A
30.6+0.7+0.65=31.95=0.65
Method B
30.8+0.85+0.9=32.55=0.85
(b) Comparison

Method	Average IoU
A	**0.65**
B	**0.85**

Higher IoU means better overlap between the predicted and actual bounding boxes.
Conclusion

Method B is better because its average IoU is **0.85**, compared with **0.65** for Method A. Therefore, Method B provides **better detection and more accurate bounding-box localization**. ✅